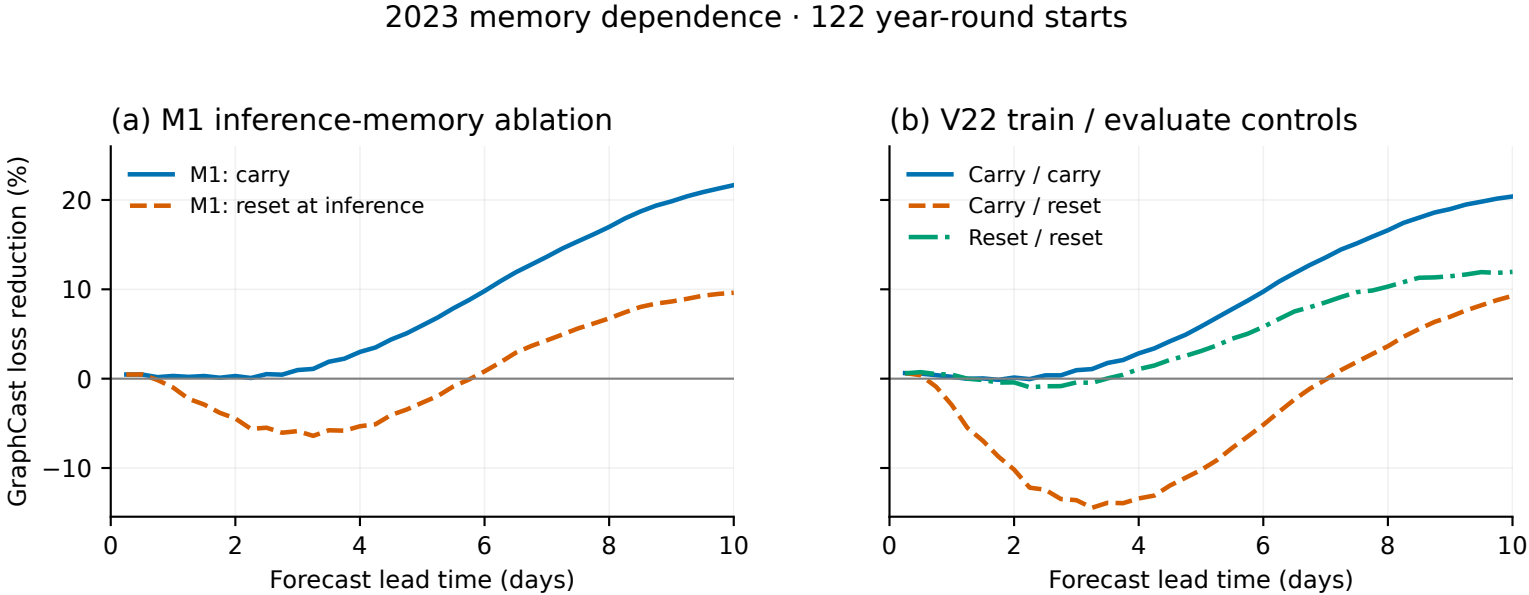
2023 memory dependence · 122 year-round starts
(a) M1 inference-memory ablation
M1: carry
M1: reset at inference
GraphCast loss reduction (%)
20
10
0
−10
0
2
4
6
8
10
Forecast lead time (days)
(b) V22 train / evaluate controls
Carry / carry
Carry / reset
Reset / reset
0
2
4
6
8
10
Forecast lead time (days)
Exact original GraphCast loss throughout; common canonical baseline.
V22 controls are separate training runs; they are not matched capacity controls for M1.